Climate change describes a change in the average conditions ? such as temperature and rainfall ? in a region over a long period of time. Global climate change refers to the average long-term changes over the entire Earth. These include warming temperatures and changes in precipitation, as well as the effects of Earth's warming, such as:

- Rising sea levels
- Shrinking mountain glaciers
- Ice melting at a faster rate than usual in Greenland, Antarctica and the Arctic
- Changes in flower and plant blooming times.

Earth's climate has constantly been changing, even long before humans came into the picture. However, scientists have observed unusual changes recently. For example, Earth's average temperature has been increasing much more quickly than they would expect over the past 150 years.

Causes of Climate Change:

Many factors, both natural and human, can cause changes in Earth's climate.

Natural causes include changes in Earth's orbit and the amount of energy coming from the sun. Ocean changes and volcanic eruptions are also natural causes of climate change.

Human activities contribute significantly to climate change by causing changes in Earth's atmosphere in the amounts of greenhouse gases, aerosols, and cloudiness. The largest known contribution comes from the burning of fossil fuels, which releases carbon dioxide gas to the atmosphere. Greenhouse gases and aerosols affect climate by altering incoming solar radiation and outgoing infrared (thermal) radiation that are part of Earth's energy balance. Changing the atmospheric abundance or properties of these gases and particles can lead to a warming or cooling of the climate system.